

SOC Investigation Report #3

Successful Login Investigation (Windows Event ID 4624)

Objective

Monitor and analyze successful Windows login events using Splunk to distinguish legitimate user and service logons from suspicious authentication activity.

Lab Environment

Component	Details
Host Machine	Windows 11
Virtualization	Oracle VirtualBox
Attacker Machine	Kali Linux
Victim Machine	Windows 10 Home
SIEM	Splunk Enterprise
Log Forwarder	Splunk Universal Forwarder
Monitoring	Windows Security Logs + Sysmon

Investigation Steps

1. Generate a successful login event by logging into Windows or observe an automatically generated service logon.
2. Open Splunk Search & Reporting.
3. Run: index=* EventCode=4624
4. Open one event and review the fields.

Key Findings

Observed Event ID: 4624
Message: An account was successfully logged on.
Account Name: SYSTEM
Account Domain: NT AUTHORITY
Logon Type: 5 (Service Logon)
Process: C:\Windows\System32\services.exe
Authentication Package: Negotiate
Source Network Address: None

Analysis

The event represents a Windows service logon, not an interactive user login. The SYSTEM account and services.exe process indicate expected operating system activity.

SOC Conclusion

This event is legitimate and does not indicate malicious behavior. No incident should be raised based on this event alone.

Splunk Queries

```
index=* EventCode=4624
```

```
index=* EventCode=4624 | stats count by Logon_Type
```

```
index=* EventCode=4624 | table _time Account_Name Logon_Type ComputerName
```

Evidence

Insert screenshots:

1. Splunk search query
2. Event details (Event ID 4624)
3. Highlighted Account Name, Logon Type, and Process Name

Skills Learned

- Windows Event ID 4624 analysis
- Logon type interpretation
- Authentication investigation
- Splunk Security Log analysis
- Basic SOC authentication investigation